Elations and two translation modules

A recovery of the even-characteristic classification in L027

Let F be a finite field of characteristic two. A string C-marking is an ordered independent generating tuple of involutions satisfying the string commutation relations and the subgroup intersection property. It is nondegenerate when every adjacent product has order greater than two. We prove the assertions of Leemans–Vandenschrick, Section 5.4–5.5, with the actual field and intersection conditions made explicit. The matrices in the normal forms are those of that source. The common geometric argument and the two-module proof below replace its separate calculations.

We retain the elementary projective interpretation of a linear action, the usual criterion which deduces the intersection property from the two end markings and their intersection, and Dickson's subgroup theorem in the following precise form. A subgroup of $\mathrm{PSL}_2(F)$ is cyclic, dihedral of order $2k$ with k odd, contained in a point stabilizer, or conjugate to a natural $\mathrm{PSL}_2(K)$ for a subfield $K \subseteq F$. These alternatives include the small groups; in particular A_5 is $\mathrm{PSL}_2(4)$, and an A_4 subgroup is in a point stabilizer. We also retain the standard faithful action of the stabilizer of a nonsingular plane conic as $\mathrm{PSL}_2(F)$. No subgroup multiplicity or source census is used.

Commutation and the flag case

Every projective involution has a unique order-two lift to $\mathrm{SL}_3(F)$, since the scalar centre has odd order dividing three. The lift has the form $I + uf$, where $u \neq 0$, $f \neq 0$ and $f(u) = 0$. Its centre is the point $\langle u \rangle$ and its axis is the line $\ker f$. If two projective involutions commute, their order-two lifts commute: an equality $AB = \zeta BA$ with scalar ζ implies $\zeta^2 = 1$.

Lemma 1. *Two distinct elations commute if and only if they have the same centre or the same axis.*

Proof. For $N = uf$ and $M = vg$, commutation is $uf(v)g = vg(u)f$. If both the vectors and the functionals are independent, this forces $f(v) = g(u) = 0$. The independent vectors u, v would then lie in the one-dimensional intersection of the distinct kernels, a contradiction. Conversely, a common centre or axis makes both products zero. □

Theorem 2. *A nondegenerate string C-marking whose group fixes a point and an incident line has rank at most three. The possibilities in ranks one, two and three are respectively C_2, D_8, and $C_2^2 \wr C_2$, of types* [], [4], *and* [4, 4].

Proof. Choose a basis along the fixed flag. Every generator is upper unitriangular. Write such a matrix as $(x, y, z) = I + xE_{12} + yE_{13} + zE_{23}$; it is an involution precisely when $xz = 0$. A generator with $x = z = 0$ is central and cannot occur in a nondegenerate marking of rank at least two. Noncommuting adjacent generators must alternate between $x \neq 0, z = 0$ and $x = 0, z \neq 0$. Their product has order four, giving D_8. At rank four the first and fourth generators are of opposite kinds and do not commute, contrary to the string relation.

At rank three, reverse the coordinates if necessary and write the ends as $r = (a, b, 0)$, $t = (e, f, 0)$, and the middle as $s = (0, c, d)$, with $a, e, d \neq 0$. Their commutators with s are

the central matrices $z_a = (0, ad, 0)$ and $z_e = (0, ed, 0)$. If $a = e$, the two dihedral ends share the nonidentity $z_a = z_e$, which is not in $\langle s \rangle$; the intersection property fails. If $a \neq e$, the four commuting involutions r, t, z_a, z_e are independent over $\mathbb{F}_2$: independence of a, e first kills the coefficients of r, t, and then that of ad, ed kills the remaining coefficients. They form a normal group of order 16. Conjugation by s sends r to rz_a and t to tz_e. It therefore interchanges the two bases (r, t) and (rz_a, tz_e), giving $C_2^2 \wr C_2$. The two dihedral end subgroups meet in $\langle s \rangle$, since their respective base spaces $\langle r, z_a \rangle$ and $\langle t, z_e \rangle$ are disjoint. This proves the intersection property and the asserted type. □

Rank three without an incident fixed flag

We write $M(x, y) = I + xE_{21} + yE_{23}$. The following small module calculation will be used twice. If $t \neq 0$, $K = \mathbb{F}_2(t)$, and the two involutions act on rows by

$$(x, y) \longmapsto (x, y + tx), \qquad (x, y) \longmapsto (x + y, y),$$

the module generated by $(0, c)$ is cK^2. Indeed the differences give $(c, 0)$ and $(0, tc)$; iteration gives every polynomial in t in both coordinates. The reverse inclusion follows from invariance. The product of the two matrices has trace t and determinant one. Its eigenvalues are λ, λ^{-1}, with $\lambda \neq 1$; its order $k > 1$ is odd, and $K = \mathbb{F}_2(\lambda + \lambda^{-1})$. Consequently k divides $|K| - 1$ or $|K| + 1$. The generated group is the actual dihedral group D_{2k}.

Theorem 3. *Let (r, s, t) be a rank-three string C-marking with no fixed incident point–line pair. Up to a projective change of coordinates and duality, its lifts have the form*

$$r = I + aE_{13}, \qquad s = I + E_{31}, \qquad t = I + bE_{13} + cE_{23}, \qquad a \neq 0.$$

These lifts fix e_2, so their group meets the scalar centre trivially. If $b = 0$ or $b = a$, then $c \neq 0$ and the group is $K^2 \rtimes D_{2k}$, where $K = \mathbb{F}_2(a)$ and the action is the natural two-dimensional dihedral action. The types are respectively $[k, 4]$ and $[k, k]$. Otherwise the group is $\mathrm{PSL}_2(K)$ for $K = \mathbb{F}_2(a, b)$.

Proof. The ends commute, so after duality they share an axis l. The axis m of s differs from l, since otherwise all three commute and fix a flag. Put $p = l \cap m$. The centre of s is not p: otherwise the whole group preserves (p, l). The centres of the two ends cannot both be p, since then (p, m) is fixed. Reversing the marking if necessary, take the centres of r, s and the point p as the three coordinate points. Scaling gives the displayed matrices.

If $b = 0$, the third generator is $M(0, c)$; if $b = a$, its product with r is $M(0, c)$. In either case $c \neq 0$ by nonidentity and independence. The group generated by r, s has trivial intersection with the translation group $M(F^2)$, and its action on that group is precisely the preceding module calculation. Thus the asserted semidirect product is an equality of actual matrix groups. When $b = 0$, the other dihedral end has order eight and intersects $\langle r, s \rangle$ in $\langle s \rangle$. When $b = a$, set $v = M(c/a, 0)$. Conjugation by v fixes s and takes r to t, so the other end is the conjugate of $D = \langle r, s \rangle$. Its intersection with D is the stabilizer in D of the nonzero vector $(c/a, 0)$. No nonidentity rotation fixes such a vector, and two distinct reflections fixing it would have a nonidentity rotation as their product. The stabilizer is therefore exactly $\langle s \rangle$. These arguments also prove the claimed types and the intersection property for these normal forms.

Now suppose $b \notin \{0, a\}$. If $c = 0$, the action on the plane $\langle e_1, e_3 \rangle$ is faithful. If $c \neq 0$, direct substitution shows that all three matrices preserve the nonsingular conic

$$Q = \frac{x_1^2 + ax_1x_3 + ax_3^2}{b(a + b)} + \frac{x_2^2}{c^2} = 0.$$

Its polar radical is $\langle e_2 \rangle$ and $Q(e_2) \neq 0$. The induced action on the quotient by this point is faithful: a kernel element has the form $M(x, y)$, and preservation of Q gives $(xx_1 + yx_3)^2/c^2 = 0$, hence $x = y = 0$. In either case the group is isomorphic to

$$H = \left\langle \begin{pmatrix} 1 & a \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix} \right\rangle \leq \mathrm{SL}_2(K), \qquad K = \mathbb{F}_2(a, b).$$

The first and third matrices are distinct commuting involutions, excluding the odd dihedral alternative. The first and second have different unique fixed points, excluding a point stabilizer. Dickson's theorem therefore gives a conjugate of $\mathrm{PSL}_2(K')$. The traces of the products of the first and second and of the second and third matrices are a, b. Conjugacy preserves trace, so $K \subseteq K'$; the group order bound inside $\mathrm{SL}_2(K)$ gives $|K'| \leq |K|$. Thus $K' = K$. The conclusion is a classification of groups of the stated C-markings; it does not assert that every remaining pair a, b has the intersection property. □

Rank four and the actual intersection

Theorem 4. *A nondegenerate rank-four string C-marking in* $\mathrm{PSL}_3(F)$ *is, up to coordinates and duality, of the form*

$$r_0 = I + E_{23}, \quad r_1 = I + aE_{31}, \quad r_2 = I + bE_{13}, \quad r_3 = I + E_{21}, \qquad ab \neq 0.$$

Put $K = \mathbb{F}_2(ab)$ *and let* k *be the order of* $r_1 r_2$*. These matrices satisfy the intersection property exactly when* $a \notin K$*, equivalently* $b \notin K$*. In that case their group is* $K^4 \rtimes D_{2k}$*, their two ends are* $K^2 \rtimes D_{2k}$*, and their type is* $[4, k, 4]$*. There is no nondegenerate marking of rank greater than four.*

Proof. The group cannot fix an incident flag by Theorem 2. After duality, r_0, r_2 share an axis l. If r_1, r_3 share a centre p on l, the whole group fixes (p, l); if their common centre is off l, then r_0, r_3 have neither a common axis nor a common centre and cannot commute. Thus r_1, r_3 share an axis $m \neq l$. Since r_0, r_3 commute, their common centre is $p = l \cap m$. Nondegeneracy forces the centres of r_1, r_2 to differ from p. Taking these three centres as coordinate points and scaling gives the displayed form.

Let $D = \langle r_1, r_2 \rangle$. On the coordinates (x_1, x_3) it is the dihedral group of order $2k$, with rotation trace ab; it meets $M(F^2)$ trivially. On this translation group conjugation is

$$(x, y)^{r_1} = (x + ay, y), \qquad (x, y)^{r_2} = (x, y + bx).$$

The module generated by $r_0 = M(0, 1)$ is $N_0 = M(aK, K)$, and that generated by $r_3 = M(1, 0)$ is $N_3 = M(K, bK)$. To see equality, successive differences first give the other coordinate and then multiplication by ab; iteration gives all polynomials in ab. The displayed spaces are invariant, proving both inclusions. Hence the two actual end groups are $N_0 \rtimes D$ and $N_3 \rtimes D$.

If $a \in K$, then $b \in K$ and $N_0 = N_3$, so the end intersection is larger than D and IP fails. Otherwise $aK \cap K = 0$ and $bK \cap K = 0$, giving $N_0 \cap N_3 = 0$ and end intersection exactly D. Each end has rank-three IP: for example $\langle r_0, r_1 \rangle$ has translation space $\mathrm{span}_{\mathbb{F}_2}\{(0, 1), (a, 0)\}$ and meets D in $\langle r_1 \rangle$. Its outer adjacent order is four, and the other end follows by reversal. The end criterion proves full IP. The sum $N_0 \oplus N_3$ has dimension four over K, which gives the group and type claimed.

Finally a nondegenerate marking of rank at least five has a rank-four initial segment of this form. Its fifth generator commutes with r_0 and r_2, which have a common axis and distinct centres. By Lemma 1, it must have that same axis. It also commutes with r_1, whose axis is different and whose centre is not on the common axis. This is impossible by the same lemma. □

Scope and attribution

Theorems 2, 3, and 4 recover the source's Theorems 5.4, 5.5, and 5.6, and the corresponding rank-three and rank-four rows of Table 2. In particular the notation for the elementary abelian groups refers to the exact trace fields displayed above; it does not identify an arbitrary divisor of the ambient field order with a subfield. The classification of rank-three projective-line C-markings remains its separately attributed earlier result. The present argument classifies their possible group, without repeating that census. The rank-two semisimple rows of Table 2 and the odd-characteristic classifications are outside this component. No chiral rank-six nonexistence result is asserted here: that argument still has to join the actual two regular end actions.